

## A Colorful Kaleidoscope Code

```
#Kaleidoscope.py
```

```
import random
```

```
import turtle
```

```
t = turtle.Pen()
```

```
t.speed(0)
```

```
turtle.bgcolor("black")
```

```
colors = ["red", "yellow", "blue", "green",
```

```
         "orange", "purple", "white", "gray"]
```

```
for n in range(50):
```

```
    # Generate spirals of random sizes/colors at random locations on the screen
```

```
    t.pencolor(random.choice(colors)) # Pick a random color from colors[]
```

```
    size = random.randint(10,40)    # Pick a random spiral size from 10 to 40
```

```
# Generate a random (x,y) location on the screen
```

```
x = random.randrange(0,turtle.window_width()//2)
```

```
y = random.randrange(0,turtle.window_height()//2)
```

```
# First spiral
```

```
t.penup()
```

```
t.setpos(x,y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.left(91)
```

```
# Second spiral
```

```
t.penup()
```

```
t.setpos(-x,y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.left(91)
```

```
# Third spiral
```

```
t.penup()
```

```
t.setpos(-x,-y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.left(91)
```

```
# Fourth spiral
```

```
t.penup()
```

```
t.setpos(x,-y)
```

```
t.pendown()
```

```
for m in range(size):
```

```
    t.forward(m*2)
```

```
    t.left(91)
```